

For any assignment, submit:

1. GitHub repo or ZIP
2. README.md with setup steps
3. Architecture diagram
4. Required config/YAML files
5. Screenshots or command outputs as proof
6. Known issues or assumptions
7. What you would improve for production

Assignment : EKS Observability Setup with Prometheus and Grafana

Objective

Set up metrics and dashboards for applications running on EKS.

Setup Requirement

Deploy a sample backend application and configure Prometheus + Grafana monitoring.

What You Need to Build

Metrics Server

Prometheus

Grafana

App Deployment

Pod CPU Dashboard

Pod Memory Dashboard

Pod Restart Dashboard

Health Check Alert

Tricky Parts

1. Metrics should be visible per namespace.
2. Dashboard should show pod CPU and memory.
3. Dashboard should show pod restart count.
4. Create one alert for app down or pod restart.
5. Simulate one failure and show it in Grafana.
6. Candidate should understand difference between logs and metrics.

Expected Proof

`kubectl top nodes`

`kubectl top pods -n devops-assignment`

`kubectl get pods -n monitoring`

Grafana screenshots:

Pod CPU

Pod Memory

Pod Restart Count

Alert Rule

Evaluation Focus

Monitoring setup

Prometheus basics

Grafana dashboards

Alerting

Failure simulation

Operational thinking